

WEIJIE HUANG

🏆 **Silver Medalist:** 2019 ICPC Asia Regional (Yinchuan)

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🎓 EDUCATION

Indiana University Bloomington

Bloomington, IN, United States

Jan. 2025 – (Expected) Aug. 2027

Ph.D. student in Computer Science

Advisor: Prof. XiaoFeng Wang, Prof. Haixu Tang, Prof. Chenghong Wang

Rice University

Houston, TX, United States

Aug. 2022 – Nov. 2024

Doctoral studies in Computer Science

Southern University of Science and Technology (SUSTech)

Shenzhen, Guangdong, China

Sept. 2018 – Jul. 2022

B.Eng. in Computer Science and Engineering

🔍 RESEARCH INTERESTS

Systems Security; Confidential Computing; GPU/AI Systems; LLM Security; Program Analysis

💻 SKILLS

- Languages: C, C++, Rust, Python, Shell, Assembly (x86, RISC-V)
- Instrumentation & Analysis tools: Intel Pin, LLVM
- AI Infrastructure & Systems: CUDA, PyTorch, Nsight, vLLM

⚙️ RESEARCH PROJECTS

High-performance Confidential Oblivious Data Analytics

Jan. 2025 – Present

Proposed efficient oblivious analytics pipelines in heterogeneous confidential computing environments to mitigate memory access pattern leakage with accelerators' on-package HBM.

1. *First author*, SP '27; 🌐 *Open-sourced*.

Designed a novel *oblivious late materialization* (OLM) workflow for relational analytics in CPU-GPU-TEE. Obtained $>100\times$ speedup on join-intensive workloads over state-of-the-art oblivious databases (Oblivator, USENIX Security '25) on TPC-H SF50.

2. *Co-first author*, CCS '26; 🌐 *Open-sourced*.

Designed a novel storage layout to fully utilize GPU HBM for efficient semantic search. Obtained up to $99\times$ latency improvements over the state-of-the-art oblivious semantic search engine (Compass, OSDI '25) at matched recall rates.

Both systems were implemented with C++/CUDA and evaluated on TDX with NVIDIA H100 (confidential computing mode enabled) across relational and vector-search workloads.

Topic-inference Attack for Realistic MoE Serving

Jan. 2026 – Present

Co-first author, *manuscript under review*.

- Proposed a simple yet effective low-privilege topic-inference attack in shared MoE serving environments. No runtime modification, instrumentation or privileged access is required throughout the attack process.
- Achieved $>98.8\%$ macro-AUROC on a multi-GPU vLLM MoE deployment with 23 candidate topic classes. Ablations showed that the attack remains effective under noisy-prompt interferences.

Automated LLM-assisted Code Optimization

July 2025 – Jan 2026

Manuscript under review; Preprint available at [arXiv](#);  Open-sourced.

- Built an open-source LLM agent for automated code optimization using trajectory-conditioned *in-context reinforcement learning* and learning-based experience reuse.
- Applied the framework to GPU-oriented scientific-computing workloads, achieving 75–300× speedups over existing implementations.

Intra-process Monitoring and Compartmentalization

Aug. 2022 – Nov. 2024

 Open-sourced.

- Extended the lab's prior *object-encapsulation analysis* (SPW '23) into a runtime monitoring and enforcement prototype using Intel Pin and LLVM.
- Built a DWARF-based policy-generation pipeline to translate static object-encapsulation results into runtime access-control policies.
- Evaluated the runtime enforcement prototype on Nginx 1.23.1 with ApacheBench across 1 KiB–4 MiB payloads, quantifying throughput overhead for instrumentation and for policy enforcement.

Trusted Execution Environment on RISC-V

May 2021 – June 2022

Co-first author, [HASP '21](#);  Open-sourced.

Proposed a flexible, software-based TEE architecture on RISC-V with novel memory management for enclaves. This architecture uses the limited standard PMP hardware to support a theoretically unbounded number of concurrent active enclaves, while prior work at the time could only support at most 13 with customized hardware primitives.

PUBLICATIONS

*: Co-first authors

- [1] **Weijie Huang**, Chenghong Wang, XiaoFeng Wang. “Towards Practical Oblivious Relational Analytics with Heterogeneous TEE”. [IEEE Symposium on Security and Privacy \(SP '27\)](#), May, 2027.
- [2] Jianzhang Du*, **Weijie Huang***, Chenghong Wang, Nicolas Tsagareli, Yukui Luo, XiaoFeng Wang, Zhongshu Gu. “Hoss: Fast Oblivious Semantic Search with Heterogeneous GPU-CPU-TEE Architecture”. [ACM Conference on Computer and Communications Security \(CCS '26\)](#), November, 2026.
- [3] Leyi Zhao*, **Weijie Huang***, Yitong Guo, Jiang Bian, Chenghong Wang, Xuhong Zhang. “Large Language Model-Powered Evolutionary Code Optimization on a Phylogenetic Tree”. [arXiv Preprint](#), January 2026.
- [4] Fangfei Yang, Bumjin Im, **Weijie Huang**, Kelly Kaoudis, Anjo Vahldiek-Oberwagner, Chia-Che Tsai, and Nathan Dautenhahn. “Endokernel: A Thread Safe Monitor for Lightweight Subprocess Isolation”. [USENIX Security Symposium \(USENIX '24\)](#), August, 2024.
- [5] Fangfei Yang, **Weijie Huang**, Kelly Kaoudis, Anjo Vahldiek-Oberwagner, and Nathan Dautenhahn. “Endoprocess: Programmable and Extensible Subprocess Isolation”. [New Security Paradigms Workshop \(NSPW '23\)](#), September, 2023.
- [6] Yudi Yang, **Weijie Huang**, Kelly Kaoudis, and Nathan Dautenhahn. “Whole-Program Privilege and Compartmentalization Analysis with the Object-Encapsulation Model”. [IEEE Security and Privacy Workshops \(SPW '23\)](#), May, 2023.
- [7] Haonan Li*, **Weijie Huang***, Mingde Ren, Hongyi Lu, Zhenyu Ning, Heming Cui, and Fengwei Zhang. “A Novel Memory Management for RISC-V Enclaves”. [Hardware and Architectural Support for Security and Privacy Workshop \(HASP '21\)](#), October, 2021.

WORK EXPERIENCE

Summer Intern (Backend)

June 2021 – Aug. 2021

Dept. Data ByteDance Ltd., Shenzhen, China

- Built an observability pipeline in Python aggregating service deployment and runtime metrics from MySQL, Grafana, and internal web sources, including CPU, storage, and network utilization.
- Deployed a scheduled FaaS workload monitor to analyze cluster utilization daily, detect overloaded services, and automatically alert service owners through Lark.
- Maintained stable operation for 2+ weeks before internship completion.